

Problem

Develop the state equations for the following differential equation.

$$\frac{d^3 y(t)}{dt^3} + \frac{6 d^2 y(t)}{dt^2} + \frac{11 dy(t)}{dt} + 6y(t) = z(t)$$

Step-by-step solution

Step 1 of 1

$$\text{Let } X_1 = y(t) \quad X_2 = X_1' \quad X_3 = X_2'$$

Thus

$$X_3' = -6X_1 - 11X_2 - 6X_3 + Z(t)$$

We can now write our state equations

$$\begin{bmatrix} X_1' \\ X_2' \\ X_3' \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} Z(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} Z(t)$$